Dust Storm Forecasting over Saudi Arabia

2026-09-22 · @Someone

A graph neural network that forecasts observed dust events at Saudi airports 3 to 48 hours ahead, by simulating dust emission and wind-driven transport across a mesh of atmospheric control volumes and reading out a severity grade at station locations. It is verified against 3.56 million station observations rather than against a reanalysis field, beats every control we have built, and is not yet an operational warning system.

The architecture

`DustTransportGNN` is a dust simulator written as a graph network. It does not classify images of the atmosphere; it emits dust, moves it along the wind, and reads off what arrives.

The graph

The domain is a 25 × 31 grid at 0.75° covering Saudi Arabia, of which **251 cells are valid land**. Each cell is a node. Edges connect the 8 surrounding neighbours, so a wind blowing at 45° is representable in a single hop rather than as two orthogonal steps.

Node positions are stored in a geographically oriented frame — `+east`, `+north` — which matters more than it sounds. Latitude descends with row index in ERA5 and CAMS, so a naive row-wise gradient is the *southward* derivative. That sign error is silent: the model still trains and the dust simply drifts the wrong way. Eight checks in `verify_dust_physics.py` pin the convention.

PhysicsFluxLayer: transport as message passing

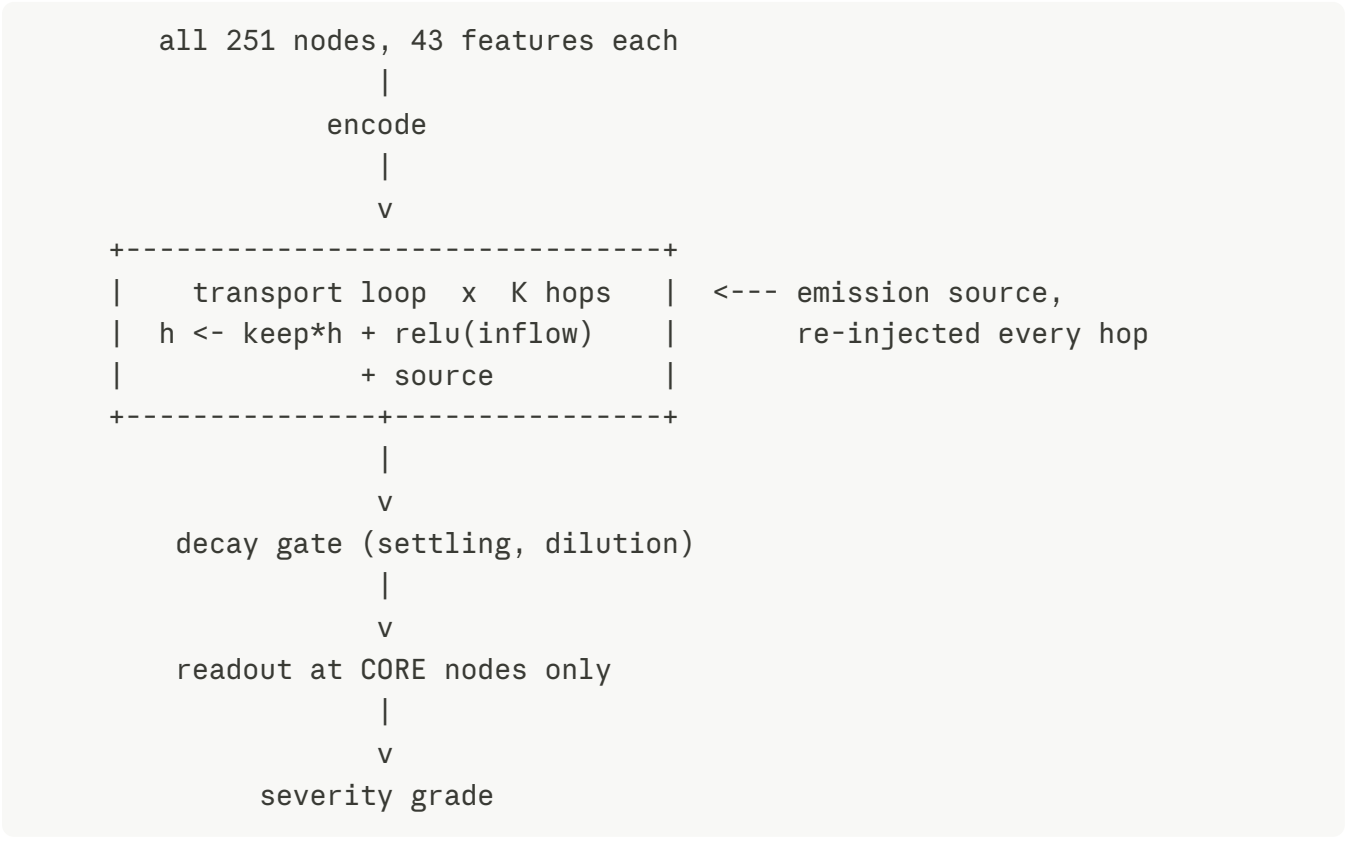
Each directed edge from source `j` to target `i` carries a message built from four things: both nodes' feature vectors, the projection of the **source cell's physical wind** onto the edge direction, and an explicit upwind advective flux

```
F[j→i]  =  ( u[j] · n[j→i] )  ×  ( C[j] − C[i] )
```

where `u[j]` is the physical wind vector at the source cell, `n[j→i]` the unit vector along the edge, and `C` the log concentration. The first bracket is the wind projected onto the edge; the second is the concentration drop across it.

Messages are summed, so each node accumulates **net inflow**. The directional prior is built into the message rather than learned from scratch: the network is not asked to discover that dust moves downwind, only how much of it survives the trip.

Emit, advect, read out



One hop is `h ← keep · h + relu(inflow) + source`. Three design choices sit in that line:

- **Every node emits**, not just the source region. Dust is lifted continuously along the transport path, so a single injection at the origin would model a puff released at t=0. Re-injecting the source term at every hop models a line of sources under a moving wind field — which is what a shamal is.
- **One shared transport operator**, applied repeatedly. The physics of a hop does not change with the hop index, and sharing keeps the parameter count flat as the horizon grows.
- **A learned decay gate** `keep = sigmoid(decay)` in (0,1), representing settling and dilution over one hop.

Hops are scaled to the forecast horizon

The single most consequential fix in the model's history. The architecture inherited `hops=2` from a 3-hour experiment, which gives a **152 km receptive field** — fine at 3 h, badly wrong at 24 h. Hops are now derived from how far dust actually travels:

Horizon	Dust travels (p90 wind)	Hops	Reach
3 h	63 km	2	152 km

Horizon	Dust travels (p90 wind)	Hops	Reach
24 h	501 km	7	532 km
48 h	1,002 km	12 (capped)	912 km

At 48 h the cap binds, and the honest reading of a capped value is that the model still cannot see its whole source region at that lead time. That is a domain problem, not a tuning problem.

The head

The model previously carried three independent sigmoid outputs, one per visibility threshold. Nothing enforced that they nest, and the rarest starved — at a 0.070% base rate the `severe` head never fired.

It now predicts a single **ordinal severity grade** trained with the SORD loss:

Grade	Condition
0 clear	visibility ≥ 5 km
1 dusty	1 ≤ visibility < 5 km
2 dust_storm	0.4 ≤ visibility < 1 km
3 severe	visibility < 0.4 km

The three threshold probabilities come back as tail sums of one softmax, so `P(severe) ≤ P(dust_storm) ≤ P(dusty)` holds by construction — verified across 500 random outputs. SORD replaces the one-hot target with a distribution decaying in ordinal distance, so forecasting *clear* for a severe storm costs more than forecasting *storm*. That is the right shape for a warning system, where under-forecasting is the failure that matters.

The rings: predict the core, feed it from the boundary

This is the design idea that most distinguishes the model, and it came out of a measurement rather than a preference.

The problem it solves

A cell at the edge of the domain has a truncated stencil — some of its 8 neighbours do not exist — and, more importantly, **no inflow**. Whatever dust is about to arrive there is currently outside the grid. Asking the model to predict such a cell is asking it to invent information.

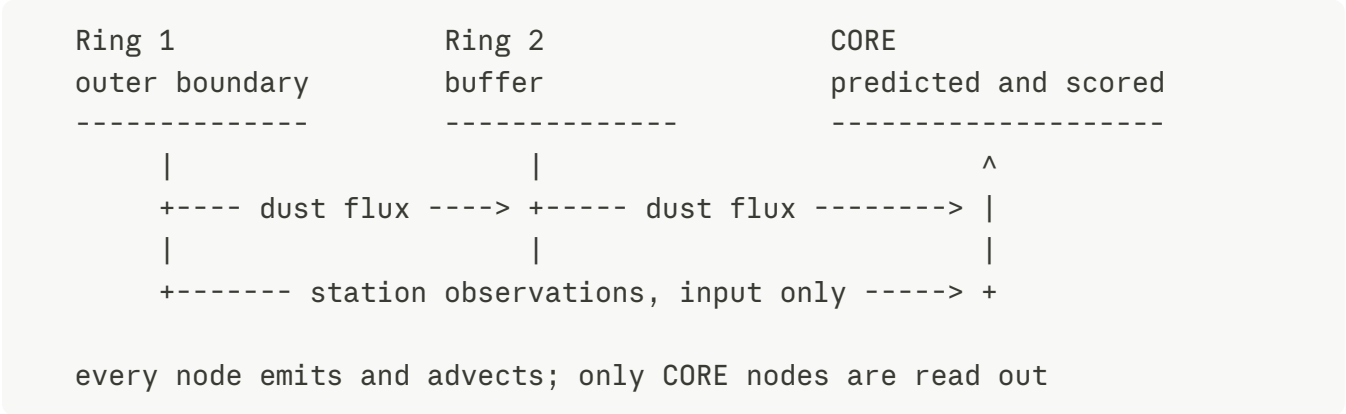
But the same cells are where imported dust *arrives first*, and imports dominate:

- **42% of observed dust storms have no local emission** — friction velocity is below the threshold for 1,409 of 3,369 storm hours.
- **68% arrive on a northerly wind.**
- An interior storm is preceded by a **northern** station's storm **48.8%** of the time within 24 h, against **21.8%** from the south. The lead is directional transport, not widespread dustiness.

So boundary cells are poor targets and excellent sources. An earlier version dropped them entirely, which was correct as targets and wrong as inputs — it discarded that 48.8% upwind lead.

How the rings work

Cells are labelled by **erosion depth**: how many successive 8-neighbour erosions from the domain edge a cell survives. Depth 1 is the outer boundary, higher is deeper inside.



`--buffer-rings N` reserves the outer N rings as **input-only**. Every node still emits, advects and participates in message passing; only core nodes are read out, and the loss is masked to them. At `--buffer-rings 2` on the current domain:

	Count
Valid land cells	251
Cells with a full 8-neighbour stencil	100
Stations in the graph	26, on 21 distinct cells
Core stations — predicted and scored	**11**
Boundary stations — upwind inputs only	15

The tension, stated honestly

A deeper buffer gives the core genuine upwind inflow, and at 24 h dust crosses about 7 cells, so the buffer *should* be deep. But every ring added converts targets into inputs. Three rings leaves **4 stations** on this domain. Two rings leaves 11.

That is the binding constraint on the whole project: **we are fitting a model on 11 labelled points.** It is why buffer depth is the right design waiting on a bigger domain, not a hyperparameter to tune here.

Observation assimilation, and the generalisation test built into it

Core stations can optionally be fed their own current visibility as three extra channels — log extinction, a has-observation flag, and the dust weather code — written at station cells and left zero elsewhere. During training each station's observation is hidden with probability 0.5, which forces the network to keep a working gridded-only pathway; otherwise it would lean entirely on the observation and be useless at the 225 cells that have none.

Every result is therefore scored twice, `obs` and `no-obs`. The gap between them is the cost of generalising to a cell with no station, and it is **0.007 TS at 24 h** — small enough that the gridded pathway is genuinely working.

The features

29 channels per node today, 32 with observation assimilation, 43 once the pressure-level block lands.

Group	Channels	Why
Current state	`log_pm10`, `log_pm25`, `u`, `v`, `speed`, `t2m`, `rh`, `log_tp`	Where the dust and the weather are now. The first three are also read *unnormalised* by the flux layer, which needs physical wind and concentration
History, 2 frames	`log_pm10_-1/-2`, `log_pm25_-1/-2`, `u_-1/-2`, `v_-1/-2`	Lets the model perceive the tendency dC/dt rather than copying the last frame
Meteorology at the valid time	`u_+1`, `v_+1`, `speed_+1`, `t2m_+1`, `rh_+1`, `log_tp_+1`	The single most important group. A 24 h dust forecast needs the wind at +24 h, not at +3 h

Group	Channels	Why
Emission	`ustar_+1`, `log_emission_+1`, `emis_tendency_+1`	Shao saltation with Fécan moisture correction, gated by a MAIAC-derived erodibility map
Terrain and transport	`rel_elev`, `transport`	Relative elevation, and the explicit advection tendency
Season	`doy_sin`, `doy_cos`	Cyclic day-of-year. Shamal season is real and the model should know the date
Assimilation, optional	log extinction, has-observation flag, dust weather code	Current station state, dropout 0.5 during training
Stability, pending download	`t_500`, `t_700_500`, `ri_850_700`, `ri_900_850`, `theta_900`, `dtheta_900_700`, `shear_900_700`, `llj_speed`, `llj_level`, `q_900`, `z_500`	The missing half of the physics — see below

Why the meteorology is drawn at the valid time

A dust forecast follows weather that a separate NWP system forecasts. An earlier version shared one model across horizons and supplied wind only at +3 h, so the 24 h head had to predict a storm a day out while seeing the wind three hours after initialisation. It scored **TS 0.025**. Feeding ERA5 at `t+h` fixed it and is the same idealisation AI-DUST's evaluation makes — it stands in for a perfect wind forecast, which flatters us relative to operations and should be said whenever these numbers are quoted.

The stability block, and why it is the biggest known gap

Every feature above is a **surface field**. The two studies closest to this problem both use five pressure levels, and Xu et al. 2026 — physics-guided ML for dust visibility over the Kumtag Desert — put pressure-level quantities in their **top three SHAP predictors at every forecast horizon**, with vertical mixing ranked ahead of both transport and emission.

The physical argument is straightforward. Emission decides how much dust leaves the ground and transport decides where it goes, but neither decides how much of it is at eye level. That is set by how deeply the boundary layer mixes: the same dust load gives 20 km visibility when stirred through 3 km of atmosphere and 500 m visibility when trapped under an inversion.

`src/physics/stability.py` derives eleven channels from ERA5 at 250/500/700/850/900 hPa — the level set from Sarafian et al. 2023. The core quantity is the bulk gradient Richardson number,

$$Ri = \frac{g \cdot (\theta_{upper} - \theta_{lower}) \cdot \Delta z}{\theta_{mean} \cdot [\, (\Delta u)^2 + (\Delta v)^2 \,]}$$

where θ is potential temperature, Δz the layer thickness from geopotential, and Δu, Δv the wind difference across the layer. The numerator is buoyant suppression of turbulence, the denominator its shear production.

which has a pole where shear vanishes. The denominator is floored at 0.5 m/s across the layer — below ERA5's own useful resolution — and the result squashed through `arctan`, which is monotone, so any threshold the model learns corresponds to a real threshold in Ri. Six checks verify the sign conventions: an inverted Richardson number is as silent a bug as an inverted advection sign.

This may also explain our own finding that **graph transport carries only 3% of feature importance**. A transport model with no stability information is being asked to explain visibility with the wrong half of the physics.

The data

Asset	State	Role
CAMS Saudi, 0.75°, 22 years	445 MB, complete	PM10/PM2.5 as **auxiliary input**, no longer the target
ERA5 emission drivers, 22/22 years	367 MB, complete	Friction velocity, soil moisture, boundary-layer height
METAR Saudi, 26 stations	142 MB, **3.56 M observations**	The target
METAR wide, 224 stations	941 MB, 23.6 M observations	24 upwind stations — Iraq 14, Syria 5, Jordan 3, Kuwait 2
MAIAC AOD, 1,210 granules	14 GB	Used only to build a static erodibility map
Erodibility map	28 KB, spread 0.286	Built and wired, never trained with
CAMS wide	20/22 years	For the wide domain
ERA5 wide	barely started	For the wide domain

Asset	State	Role
ERA5 pressure levels	**downloading**	The stability block

The target, and why it changed

The project trained against CAMS PM10 for its first year. Validated against **234,512 matched station-hours**, that field under-represents real dust storms by **15–49×**: across 3,369 dust-coded storm hours CAMS reported a median of 268 µg/m³ where the observed visibility requires 3,912–13,040 µg/m³ by the Koschmieder relation, and it never once reached the lower bound. The control that makes this a finding rather than a complaint: **fog hours came back at 48 µg/m³**, correctly clean. The field is not uniformly biased — it fails specifically on dust mass.

> **Flagged, not resolved.** Two versions of this figure are live in the repo. `HANDOVER.md` and `literature.py` say 15–49× over 3,369 hours; `station_target.py` and `train_station_dust.py` say 17–56× over 499 hours. The denominators differ sevenfold, so these are plausibly different subsets — any dust weather code, against those coded specifically as a duststorm — but that is unconfirmed. One should be retired or the distinction written down. Both put CAMS one to two orders of magnitude low.

Airport visibility does contain dust storms, is observed rather than modelled, and there is a great deal of it. Visibility is converted to extinction by Koschmieder, `β = 3.912 / V`, because extinction is approximately linear in coarse-mode dust mass whereas visibility is proportional to its reciprocal and saturates. METAR reports up to a 9.5 km ceiling, so clear air is right-censored in visibility — but every grade boundary sits far below the ceiling, so a censored observation is unambiguously grade 0 and censoring does not bite on the ordinal target.

Splits and label counts

Strictly chronological, no leakage. Normalisation statistics are fitted on the training split only and stored as model buffers.

Split	Years	Samples
Train	2003–2018	46,749
Validation	2019–2020	5,847
Test	2021–2024	11,671

Base rates, measured over daily dust-days at the 11 core stations:

Grade	Base rate	Events in test
`dusty`, vis < 5 km	**8.09%**	975
`dust_storm`, vis < 1 km	**1.26%**	152
`severe`, vis < 0.4 km	**0.23%**	**28**

Twenty-eight severe events cannot separate two models. That grade is better described as unmeasurable than as hard.

Two negative data findings worth keeping

SoilGrids is useless over Arabia — 0 of 20 cells above 80% sand, total spread 22 percentage points, in a country of sand seas. **Measured ERA5 emission drivers changed nothing** — six horizon/threshold pairs all within ±0.025 noise despite the emission-active fraction moving 5.1% → 19.5%. Better emission physics is not the bottleneck.

How a forecast is made

One model per horizon, each fed meteorology at its own target time.

1. **Assemble the state.** For every one of the 251 land cells, build the 29-channel vector: current dust and weather, two history frames, the forecast meteorology at `t+h`, the emission diagnostics, terrain, and the season encoding.
2. **Assimilate what is observed.** At the 26 cells holding a station, write the current visibility, a has-observation flag and the dust weather code. Everywhere else these stay zero, and the flag tells the network the difference between "no dust reported" and "no station here".
3. **Encode and emit.** Each node is encoded to a hidden state, and a source term is computed from the emission channels.
4. **Advect for K hops.** K is set by the horizon — 2, 7 or 12. Each hop moves dust along wind-projected edges, re-injects the local source, and applies the learned decay gate.

5. **Read out the core.** Only cells inside the buffer rings produce an output. The network emits four ordinal logits; tail sums of their softmax give `P(vis < 5 km)`, `P(vis < 1 km)` and `P(vis < 0.4 km)`, monotone by construction.
6. **Aggregate to dust-days.** A station has a dust day if any hour within the calendar day crosses the threshold; the day's forecast is its peak probability. This is the unit WMO SDS-WAS and AI-DUST report, and scoring 3-hourly station-hours instead makes the event 4.5× rarer and not comparable to published scores.
7. **Threshold.** Currently frequency-matched — the operating threshold is set so the forecast fires as often as the event occurs. This is a scoring convention, not a deployment choice, and the operating-point section below shows why that distinction matters.

Training uses class-weighted SORD, Adam, gradient clipping at 1.0, dropout 0.3, early stopping on validation loss with patience 12. The model is deliberately small — hidden width 64 — because with 11 supervised nodes the first version overfit hard, with validation bottoming at epoch 5 and then rising 20%.

On a GTX 1650 an epoch takes about 1.1 min at 3 h, 2.8 min at 24 h, and longer at 48 h where the hop count reaches 12.

What the current model achieves

Every number below is scored against **observed station visibility**, daily dust-days, core stations only, test period 2021–24. `no-obs` withholds the station's own current reading and is the column comparable to published forecast scores.

Horizon	Grade	Mode	TS	ETS	POD	FAR	Events
3 h	dusty	no-obs	0.296	0.263	0.457	0.543	1,230
3 h	dust_storm	no-obs	0.175	0.170	0.297	0.703	185
3 h	severe	no-obs	0.029	0.028	0.057	0.943	35
24 h	**dusty**	**no-obs**	**0.267**	**0.234**	**0.421**	**0.579**	**1,229**
24 h	dust_storm	no-obs	0.142	0.137	0.249	0.751	185
24 h	severe	no-obs	0.000	−0.001	0.000	1.000	35
48 h	dusty	no-obs	0.204	0.170	0.339	0.661	1,226
48 h	dust_storm	no-obs	0.085	0.080	0.157	0.843	185
48 h	severe	no-obs	0.045	0.044	0.086	0.914	35

Against every control

At 24 h, `dust_storm` grade, identical data:

Control	TS	ETS	POD	FAR
never-dust	0.000	0.000	0.000	—
climatology, station × month	0.039	0.033	0.092	0.937
wind threshold at +24 h	0.072	0.067	0.135	0.865
logistic regression	0.072	0.067	0.135	0.865
persistence	0.076	0.070	0.141	0.859
gradient boosting	0.098	0.093	0.178	0.822
flux-GNN	**0.142**	**0.137**	**0.249**	**0.751**

1.9× persistence, 3.6× climatology, 1.45× gradient boosting. A wind threshold alone gets 0.072, so roughly half the model's skill is "it will be windy" and half is everything else.

Read SEDI down, TS across

TS, ETS, POD and FAR all fall with the base rate whether or not the forecast got worse. `verify_metrics.py` demonstrates it: holding forecast quality fixed at H = 0.42, F = 0.03 and varying only how often the event happens,

Grade	Base rate	TS	ETS	SEDI
dusty	7.0%	0.300	0.270	0.637
dust_storm	1.0%	0.106	0.098	0.637
severe	0.2%	0.026	0.024	0.637

TS falls **11.4×**; SEDI does not move. So TS compares models at one threshold and nothing else, and **SEDI** — base-rate independent by construction — is what to quote across grades. This also retires an earlier conclusion: ETS looked near-redundant at an 8.6% base rate, but it collapses just as TS does at 0.2%.

The degenerate case is real, not hypothetical: `severe` at 24 h scores TS 0.000, which reads like a measurement, against SEDI NaN, which correctly says the contingency table is empty.

What the model actually uses

Permutation importance at 24 h, `dust_storm`, against a baseline average precision of 0.0617:

Channel	AP drop
`speed_+1`	**+0.0288**
`log_pm25`	+0.0158
`emis_tendency_+1`	+0.0157
`rh_+1`	+0.0138
`ustar_+1`	+0.0121
`v_+1`	+0.0115
`doy_cos`	+0.0099
`transport`	+0.0062
`speed` (current)	**−0.0124**

Wind speed at the valid time alone is **nearly half the model**. Grouped: future meteorology 40% plus emission 18% is about 58%; CAMS PM10 **5%**; graph transport **3%**. The current-time `speed` channel has *negative* importance — permuting it helps — and should simply be dropped.

No usable operating point yet

Horizon	Target POD	FAR	TS	Alerts per station per year
3 h	0.5	0.892	0.098	22
24 h	0.5	0.948	0.049	46
24 h	0.7	0.971	0.029	115
24 h	0.9	0.992	0.008	511
48 h	0.5	0.942	0.055	41

Catching half the dust storms a day ahead means 46 alerts per station per year, 19 in 20 of them wrong. One qualification: that verdict assumes a symmetric cost, and nobody has yet computed the cost–loss ratio at which this has positive value for a named user.

Our numbers against the literature

The comparison has to be matched on base rate

This is the single most important methodological point in the report. A threat score is only meaningful beside another threat score measured on an event of similar rarity. Every published "dust event" threshold sits at or below our `dusty` grade:

Definition	Base rate
Sarafian 2025: PM10 > 50 µg/m³	22%
Sarafian 2023: PM10 > 65.2 µg/m³	13%
Ours, `dusty`: vis < 5 km	**8.09%**
Xu 2026: vis ⩽ 10 km	under 8%
Ours, `dust_storm`: vis < 1 km	1.26%
Ours, `severe`: vis < 0.4 km	0.23%

The project previously quoted its `dust_storm` TS of 0.135 against published numbers measured at 13–22%, and concluded it was "about 2× below the operational bar". **That claim has been withdrawn.** Read at the grade the literature actually uses, the comparable figure is 0.267.

Head to head

Study	Region	Task	Lead	Base rate	Result
Sarafian et al. 2023	Israel, 30 stations	PM10 dust event, ordinal	24 h	13%	recall **0.76**, precision **0.67**, implied **TS 0.553**
This project	**Saudi, 11 core stations**	**visibility grade, ordinal**	**24 h**	**8.09%**	**TS 0.267**, POD 0.421, FAR 0.579
Sarafian's XGBoost control	same	same	24 h	13%	recall 0.67, precision 0.47, implied TS 0.382
Sarafian's persistence control	same	same	24 h	13%	recall 0.51, precision 0.51
Xu et al. 2026	Kumtag Desert, 4 stations	visibility grades from IFS	0–24 h	<8%	QWK 0.264
AI-DUST 2025	East Asia	gridded dust, AIFS-driven	day 1	unstated	TS 0.33

Study	Region	Task	Lead	Base rate	Result
Karami et al. 2021	Middle East, 9 operational models	column AOD, 12 events	24 h	—	best R^2 0.78 (DREAM8-MACC)

Sarafian is the one genuinely comparable result, and we are behind it — TS 0.267 against an implied 0.553, at a base rate 1.6× lower. Not a small gap. But it is explainable by inputs rather than by architecture: they use five pressure levels, time-varying AOD, a Sahara-wide input domain and a dense auxiliary task. We have none of those four. Notably their *XGBoost control* at 0.382 also beats us, which is consistent with our own finding that architecture is not where the leverage is.

The Saudi literature is not competition

No published Saudi study forecasts a named station's dust event at a lead time.

Study	What it does	Why it is not the same task
Alshammari et al. 2024 — Riyadh, Jeddah, Dammam	Regresses the **count of dust days next month**; TCN R^2 0.51, LSTM Dammam R^2 0.64	Monthly counts on 192 points per city. MAE 2.80 means wrong by about three dust days per month, with no statement about which day
Al-Ahsa classifier 2026	Daily detection; RF F1 0.88, AUC 0.97	**Visibility is one of the input features**, against a visibility-derived label — close to circular. And zero lead time
Red Sea maritime risk map	POD 0.92, ETS 0.85	Risk mapping, not a lead-time forecast
AtmoStack, Buraidah	R^2 daily 0.73, monthly 0.96	Monthly R^2 is aggregation, not forecast skill

That is unoccupied ground, and it is the strongest argument for this project's novelty: a **day-resolved, station-specific, honestly-verified dust event forecast for Saudi Arabia does not currently exist in the literature.**

Where we stand against the operational industry

The operational system for this exact domain is the **WMO Sand and Dust Storm Warning Advisory and Assessment System**. Its GCC regional node, launched in 2023, is hosted by the Saudi National Center for Meteorology in Jeddah and covers Bahrain, Saudi Arabia,

Kuwait, Oman, Qatar and the UAE. It runs a multi-model ensemble — BSC-DREAM8b, DREAM8-MACC, NASA-GEOS, NMMB-BSC, NOA-WRF-Chem and others.

What the operational models actually achieve here

Measure	Value
Best model AOD correlation, 12 Middle East events, 24 h	R^2 0.78 (DREAM8-MACC), 0.74 (NASA-GEOS)
POD, Sahara / Sahel	above 0.68, FAR 0.16
POD, Middle East sub-regions	**0.19 – 0.48**
Systematic behaviour	**All nine models underestimate peak dust AOD**, worst for cyclonic storms

Our POD of 0.42 at `dusty` /24 h sits inside that 0.19–0.48 Middle East band. That is not a claim to have beaten the operational ensemble — they forecast a gridded field over a continent and we forecast a grade at eleven airports — but it does mean the project is in the right range rather than an order out.

> **A caution that must travel with this comparison.** The figure this project has been measuring itself against — "WMO SDS-WAS operational TS 0.31 at 24 h" — could not be traced to a primary source. The AEMET technical report that would settle it would not extract. Until it is sourced, 0.31 should not be quoted as the bar. The POD 0.19–0.48 range above is secondhand too and carries the same warning.

Three structural advantages we have

- **The operational models are verified against AOD and AERONET; we are verified against 3.56 million station observations of the thing users actually care about** — visibility at an airport. Column AOD and surface visibility are different quantities, and the nine-model study shows the models fail hardest exactly where warnings matter.
- **Cost.** A forecast step is milliseconds on a consumer GPU. The operational ensemble is a supercomputing workload. DustNet made the same point at continental scale: 1.3 M parameters, 7 minutes of training on a laptop, beating CAMS at 95% of grid cells.
- **Assimilation is the lever they are already pulling.** Karami's two best models are the two that assimilate MODIS. Our observation-assimilation channels are the same idea, and the measured cost of removing them is only 0.007 TS — meaning the gridded pathway stands on its own.

Where we are behind

- **Coverage.** Eleven core stations against a continental field.
- **Lead time.** Skill is gone by 48 h at the storm grade (TS 0.085). Operational products run to 72 h.
- **Deployability.** No operating point yet gives an acceptable false-alarm rate.
- **Input realism.** We feed ERA5 at the valid time, which stands in for a perfect wind forecast. The operational models run on a real NWP forecast and carry its errors.

The honest summary: **a credible research prototype in the same skill band as operational products on this domain, verified more rigorously than most of them, and not deployable.**

What the model is good for today

It is not a public warning siren. It is useful now in four narrower ways.

1. A screening layer for a human forecaster. At 24 h it catches 42% of dusty days. A duty forecaster who is shown those flags and discards the obvious false alarms using satellite imagery and synoptics is better off than one starting cold. The model's job is to say *look here today*, not *issue a warning*.

2. Tiered alerts where the cost of a false alarm is low. The right framing is a cost–loss ratio, not an accuracy target. Where acting on a warning is cheap — pre-positioning maintenance crews, moving a solar-panel wash cycle, scheduling outdoor work — 46 alerts per station per year at POD 0.5 may already pay. Where it is expensive — closing a runway — it does not. Computing that ratio for a named user is a short analysis nobody has done, and it would convert "not usable" into "usable for these users, not those".

3. Nowcasting is the strongest product. At 3 h the model reaches TS 0.296 at `dusty` and 0.175 at `dust_storm` — roughly double the 24 h storm figure. Short-range is where the skill actually lives, and it maps onto real decisions: ground-stop timing, visibility advisories, dust-event tagging for air-quality records.

4. As a measurement instrument, it has already paid for itself. The CAMS validation is a publishable negative result that stands independently of any forecast skill: a widely used reanalysis product under-represents Middle East dust storms by one to two orders of magnitude, demonstrated on 234,512 matched station-hours with a fog control isolating the deficit to dust mass. Anyone in the region training on CAMS PM10 needs to know that, and Sarafian et al. 2025 have shown the correction is tractable.

The reusable assets

Independently of the model, the project now owns: a 22-year harmonised Saudi dust dataset; a 3.56 M-observation station target pipeline with Koschmieder conversion and censoring handling; a verified physics library (emission, transport, stability) with 14 sign-convention tests; a rare-event metric suite with SEDI and 19 tests; and a literature reference set with per-entry provenance flags.

Upcoming steps

The diagnosis first

The evidence says the bottleneck is **data — missing predictors and too few labels — not architecture**:

1. Flux-GNN 0.142 against gradient boosting 0.098, while graph transport is 3% of importance. On the older CAMS target, gradient boosting *beat* the physics-informed GNN outright.
2. `speed_+1` alone is nearly half the average precision. The model has two real signals: how windy it will be, and the season.
3. Real ERA5 emission drivers moved the emission-active fraction almost fourfold and moved the scores by less than seed noise.
4. 42% of storms are imported from outside the grid — unpredictable in principle here.
5. Eleven labelled points; 28 severe events in four years.

That ranks the roadmap.

1. The wide domain — the biggest lever

The domain currently stops at the Saudi border, which is upwind of nothing. Extending it north and west over Iraq, Syria and Jordan does three things at once:

- Brings the **source region inside the grid**, making the 42% of imported storms predictable in principle rather than only detectable on arrival.
- Lets the buffer rings be **deep enough to match the physics**. At 24 h dust crosses about 7 cells; three rings is the right design and currently leaves only 4 core stations. On a wide domain a deep buffer becomes affordable.
- Unlocks the **224-station METAR archive as targets**, not just inputs — roughly a twentyfold increase in labels, against the 11 we fit today. This is the direct fix for the binding constraint.

Status: CAMS wide is at 20/22 years; ERA5 wide is barely started. This is the expensive item and the one most likely to move the number.

2. Vertical structure — in flight

The ERA5 pressure-level download is running. The CDS cost limit for this dataset is far tighter than for single levels — measured, not guessed: 1 variable × 1 quarter is accepted, 5 variables × 1 month is refused — so it is 440 requests. A first attempt at 6 concurrent workers had 15 of 20 parts rejected because the cap is on requests *queued per dataset*; it now runs 2-wide with exponential backoff. **24–30 hours.**

`experiments/ablation_levels.py` will then run base / +levels / +ordinal / both across multiple seeds, and report any effect smaller than the seed spread as noise. The measured seed spread on this project is 0.033 TS, larger than most effects it has chased.

3. The rest, ranked

Step	Rationale	Cost
Drop the `speed` channel	Negative importance — permuting it *helps*	free
Run leave-one-station-out	The `--loso` flag exists and has never been run. Spatial generalisation is currently an assumption, not a result	cheap
Train with the erodibility map	Built, wired, never used. Caveat: 122 MAIAC days, MODIS overpass ~10:30 local, so biased to daytime sources	cheap
Time-varying AOD as an input	14 GB of MAIAC used only for a static map. Sarafian's integrated gradients rank AOD among the top predictors	moderate
Regional auxiliary task	Sarafian's core trick for rare events with scarce labels: predict the regional CAMS field jointly. Gives CAMS a job it can actually do	moderate
Bias-correct CAMS	Public code; we hold a larger training set than the authors did, over a region where the bias is an order of magnitude worse	moderate
Drive from a real NWP forecast	Xu post-processes IFS, AI-DUST is driven by AIFS. Ours uses ERA5 at the valid time, which is optimistic relative to operations	moderate
A 0–6 h nowcast branch	DustCast holds skill above 0.5 to 5 h on SEVIRI; Meteosat has sat at 41.5°E since 2017	new data path

Step	Rationale	Cost
Cost–loss operating point	Converts "not usable" into a statement about a named user	analysis only
Contact the WMO GCC node, Jeddah	The operational system for this domain; holds the PME and NCM networks. Natural comparison and collaborator	an email

4. Reproducibility debt to clear before publication

Three of eleven recorded findings have **no reproducing script** — including the headline CAMS result, the 42%-imported figure, and the upwind-lead statistic. These are precisely the numbers a reviewer will ask to see re-run. Writing a script for each is cheap against the risk of being unable to reproduce the project's central claim on request. Everything in the station-verified tables above *is* backed by a retained results file.

Sources

Read in full, and the source of every published number quoted above:

- Sarafian et al., Deep multi-task learning for early warnings of dust events implemented for the Middle East, npj Clim Atmos Sci 6:23, 2023 — the one directly comparable study
- Xu et al., A physics-guided machine learning framework for enhancing dust storm visibility prediction, Sci Rep 16:11824, 2026 — physics as features, and the SHAP ranking that motivates the stability block
- Sarafian et al., Correction of CAMS PM10 reanalysis improves AI-based dust event forecast, Remote Sens 17(2):222, 2025 — the method paper for our CAMS finding
- Alshammari, Alrwais & Aksoy, Machine learning forecast of dust storm frequency in Saudi Arabia, Atmosphere 15(5):520, 2024 — the state of published Saudi work
- Karami et al., Evaluation of nine operational models in forecasting synoptic dust events in the Middle East, Geosciences 11(11):458, 2021 — what the operational ensemble achieves on our storm types
- DustNet: skillful neural network predictions of Saharan dust, arXiv:2406.11754, 2024

Cited but not opened, so their figures are leads rather than citable numbers: the Al-Ahsa classifier (paywalled), AI-DUST, DustCast, Ferro and Stephenson on extremal dependence indices, and the AEMET SDS-WAS model intercomparison report — the document that should settle the TS 0.31 question, whose PDF would not extract.

The full reference set, with per-entry base rates and provenance flags marking whether each number came from the paper's own text or only from a search result, lives in `src/metrics/literature.py`.

Key code

Module	What it holds
`src/models/dust_transport.py`	`DustTransportGNN` — emit, advect, read out; `hops_for_horizon`
`src/layers/flux.py`	`PhysicsFluxLayer` — wind-projected message passing
`src/models/ordinal.py`	SORD loss, monotone tail sums, class weighting
`src/physics/stability.py`	Richardson numbers, jet, shear from pressure levels
`src/physics/emission.py`	Shao saltation, Fécan moisture, erodibility hook
`src/data/station_target.py`	METAR → visibility → extinction, censoring
`src/metrics/aidust_metrics.py`	TS, ETS, HSS, **SEDI**, FGE, base rates
`experiments/train_station_dust.py`	Main training entry point
`experiments/ablation_levels.py`	The multi-seed ablation
`experiments/verify_dust_physics.py`	14 sign-convention checks — re-run after any change
`experiments/verify_metrics.py`	19 metric and ordinal-head checks